



A man in a autonomous driving test vehicle

to autonomous, liability of the manufacturer is more than of the drivers. Developing unsafe products simply means that manufacturers have to be legally responsible for incidents involving their products.

Building AV conducive infrastructure. Here, the government has to closely monitor the development and evolution of AVs for determining impact on road infrastructure, cities, and communities.

These important points should be addressed before deploying AVs at scale on public roads.

Safety principles and strategies at design level

Functional safety. It is the ability to detect, diagnose, and safely mitigate a faulty automotive electrical and electronics (EE) condition, preventing potential fatalities.

It can be further divided based on the diagram shown in Fig. 1.

Malfunction of EE component. It comprises five scenarios:

1. Unintended acceleration
2. Unintended deceleration
3. Unintended loss of acceleration
4. Unintended loss of deceleration
5. Unintended vehicle motion

For example, let us select unintended acceleration as a malfunction and work through the functional safety chain.

Hazard or unintended situation. Consider a situation where a car standing at a traffic light starts moving unintentionally into a crossing way, with crossing traffic, because of this malfunction.

Risk of harm/damage. After identifying the hazard, the next thing is to determine the risk factor based on severity, probability of exposure (likelihood of hazard), and controllability (how controllable the system is if a hazard occurs). External parameters such as vehicle speed, weather conditions, road conditions, and other driving situations also help determine the final risk factor.

Risk reduction. Based on the above three parameters and any external information, the automotive safety integrity level (ASIL) rating is derived. It is on a five-point scale, going all the

way from quality management to ASIL—A, B, C, D; defining the risk reduction measures for a particular function. Hence, the requirement for additional risk reduction measures increases as it goes from ASIL-A to ASIL-D.

While performing hazard analysis for the entire system, different ASIL ratings for different functions are obtained. On combining them only the highest one gets selected, which becomes the final ASIL rating of the system under consideration.

Safety of the intended functionality (SOTIF).

Functional safety sets the base for an overall safety strategy. But situations arising

from technological limitations due to sensor performance constraints or inadequate datasets are not covered by functional safety. There are many uncertainties; functional safety is just not enough. This is where SOTIF comes into the picture. SOTIF guides automotive design, verification, and validation measures for achieving safety without failure.

SOTIF can be better understood by going through the divisions depicted in Fig. 2.

Area 1 focuses on Known, Safe scenarios

Area 2 focuses on Known, Unsafe scenarios

Area 3 focuses on Unknown, Unsafe scenarios

Area 4 focuses on Unknown, Safe scenarios

SOTIF's goal is to maximise Area

1 and Area 4 for a safe system or function behaviour. It is also responsible for minimising Area 2 corresponding to known potentially dangerous behaviours. Some contributing measures for minimising are identifying where system/function improvements can



Fig. 2: Areas of different safety scenarios